

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	6 th March 2026
Team ID	
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion & Cleaning	SQL data extraction, handling missing values in health indicators, data type standardization for Tableau.
FR-2	Risk Factor Analysis	Categorization of BMI into bins, calculation of health mental/physical percentile ranks, heart disease prevalence by age group.
FR-3	Interactive Visualization	Development of dynamic filters (Smoking, Alcohol, Diabetes), creation of interactive maps/charts, implementation of drill-down features.
FR-4	Web Integration	Development of Flask backend, embedding Tableau dashboards into HTML templates, managing responsive UI layouts.
FR-5	User Insights	Benchmark comparison for individual health metrics, generation of summary risk stories for medical practitioners.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard must be intuitive for non-technical users (like patients) to interpret their risk levels using clear color coding and legends.
NFR-2	Security	Health data must be handled according to privacy standards; the Flask application must implement secure routing and data handling.
NFR-3	Reliability	The connection between the SQL database and the Tableau dashboard must maintain high uptime to ensure real-time analysis.
NFR-4	Performance	Dashboards should render and respond to filters within 2-3 seconds, even when processing the 319,000+ record dataset.
NFR-5	Availability	The web-integrated platform should be accessible via standard browsers (Chrome, Firefox, Safari) 24/7 for healthcare providers.
NFR-6	Scalability	The architecture should support the addition of newer healthcare datasets (e.g., genomic or lab-specific data) without requiring a complete redesign.